

Galois Theory / MM7043 2026W

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1. FIELDS

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Throughout the semester, we will employ the following conventions:

1. Unless otherwise specified, all rings will be assumed to be unital and commutative.
2. A *ring homomorphism* is a map of rings $\varphi : R \rightarrow S$, such that
 - i. $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in R$.
 - ii. $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, for all $x, y \in R$.
 - iii. $\varphi(1_R) = 1_S$

Definition 1.1. Let R be a ring. An element $x \in R$ is called a *unit* or *invertible*, if there exists an element $y \in R$ such that $x \cdot y = 1$.

The set of all units of R form a group under multiplication and will be denoted R^* .

Example 1.2.

1. $\mathbb{Z}^* = \{-1, 1\}$
2. $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$
3. For $n \in \mathbb{N}$, it holds that $\mathbb{Z}_n^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

Definition 1.3. A *field* is a ring K such that $K^* = K \setminus \{0\}$.

Example 1.4. We recall the following fields:

1. $\mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$
2. $\mathbb{Z}_p$, where $p \in \mathbb{P}$.

Lemma 1.5. A ring R is a field if and only if the only ideals of R are $\{0\}$ and R .

Proof. Let $I \subseteq R$ be an ideal. If I contains a unit then $I = R$.

“ $\implies$ ”: Assume R is a field. If $I \subseteq R$ is a non-zero, non-trivial ideal, then I contains a unit, which implies $I = R$.

“ $\impliedby$ ”: If R is not a field, then there exists some non-zero, non-unit element $x \in R$. But then the ideal Rx satisfies $\{0\} \subsetneq Rx \subsetneq R$. ■

Corollary 1.6. *A homomorphism of fields is always injective.*

Proof. Let $\varphi : \mathbb{K} \rightarrow \mathbb{L}$ be a homomorphism of fields. Then $\ker \varphi$ is an ideal in K . We have $1_{\mathbb{K}} \notin \ker \varphi$, therefore $\ker \varphi = \{0\}$ and φ is injective. ■

1.1. Field extensions

When constructing new fields, there are two main approaches:

1. The approach “from above”, where we consider intersections of fields that satisfy our requirements of an extension, which we will discuss in the following.
2. The approach “from below”, where we explicitly construct a new field based on roots of polynomials. We will discuss this approach in the next section.

Lemma 1.7. *Suppose $\mathbb{K}$ is a field and $\{\mathbb{F}_\alpha : \alpha \in I\}$ is a family of subfields of $\mathbb{K}$. Then*

$$\bigcap_{\alpha \in I} \mathbb{F}_\alpha$$

is a subfield of $\mathbb{K}$.

Definition 1.8. Suppose $\mathbb{L}_0 \subseteq \mathbb{L}$ are fields and $\alpha_1, \alpha_2, \dots \in \mathbb{L}$. Then we define

$$\mathbb{L}_0(\alpha_1, \alpha_2, \dots)$$

as the intersection of all subfields of $\mathbb{L}$ that contain $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$, and will be called the *extension* of $\mathbb{L}_0$ generated by $\alpha_1, \alpha_2, \dots$.

It is clear that such an extension is the smallest subfield of $\mathbb{L}$ that contains $\mathbb{L}_0$ and the elements $\alpha_1, \alpha_2, \dots$.

Example 1.9.

1. Consider the fields $\mathbb{Q} \subseteq \mathbb{R}$, then one can show

$$\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}.$$

2. Since the element $1_{\mathbb{Q}}$ generates $\mathbb{Q}$ as a field, it follows that $\mathbb{Q}$ has no proper subfields, i.e. $\mathbb{Q}$ is a *primary field*.
3. The field $\mathbb{F}_p$ is a primary field for any $p \in \mathbb{P}$.

Definition 1.10. Let $\mathbb{F}$ be a field. The *primary subfield* of $\mathbb{F}$ is the subfield generated by the element $1_{\mathbb{F}}$.

Definition 1.11. Let R be a ring, then its *characteristic* is defined by

$$\text{char } R := \begin{cases} \min\left\{n \in \mathbb{N} : \sum_{j=1}^n 1_R = 0\right\}, & \text{if such an } n \text{ exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1.12. Let $\mathbb{F}$ be a field.

1. If $\text{char } \mathbb{F} = 0$, then the primary subfield of $\mathbb{F}$ is^[1] $\mathbb{Q}$.
2. If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then the primary subfield of $\mathbb{F}$ is $\mathbb{F}_p$.

Lemma 1.13. Suppose $M \subsetneq R$ is a maximal ideal. Then the quotient R/M is a field.

Remark 1.14. Suppose $\mathbb{K}$ is a field and consider $\mathbb{K}[x]$, the ring of polynomials in x with coefficients in $\mathbb{K}$. Recall that $\mathbb{K}[x]$ is a principle ideal domain. A non-trivial ideal $(f(x))$ is maximal in $\mathbb{K}[x]$ if and only if $f(x)$ is an irreducible polynomial. In particular, $\mathbb{K}[x]/(f(x))$ is a field whenever $f(x)$ is irreducible over $\mathbb{K}$.

Moreover, $\mathbb{K}[x]/(f(x))$ contains $\mathbb{K}$ as a subfield, via the homomorphisms

$$\mathbb{K} \hookrightarrow \mathbb{K}[x] \rightarrow \mathbb{K}[x]/(f(x)).$$

Therefore, for every irreducible $f(x)$ the field $\mathbb{K}[x]/(f(x))$ is a field extension of $\mathbb{K}$.

Note that $f(x)$ being irreducible with $\deg f > 1$ implies that $f(x)$ has no roots in $\mathbb{K}$. But $f(x)$ always has a root in $\mathbb{K}[x]/(f(x))$, because $f(x)$ is zero as an element of $\mathbb{K}[x]/(f(x))$.

The above demonstrates, roughly, that there are relationships:

$$\begin{array}{ccc} \text{Polynomials} & \longleftrightarrow & \text{Field extensions} \\ \text{Properties of roots} & \longleftrightarrow & \text{Symmetries of field extensions} \end{array}$$

Practically, we want some criteria to determine if a polynomial is irreducible.

^[1]We always mean “up to isomorphism”.

Remark 1.15. If $\mathbb{K}$ is a field and $p(x) \in \mathbb{K}[x]$, $\deg p > 1$ is a polynomial with a root in $\mathbb{K}$, then $p(x)$ is not irreducible. Indeed, for any $a \in \mathbb{K}$ one can write

$$p(x) = q(x)(x - a) + p(a)$$

for some polynomial $q(x) \in \mathbb{K}[x]$. If a is a root of $p(x)$, this demonstrates that $p(x)$ is not irreducible.

Note that the converse fails: The polynomial $(x^2 + 1)^2$ has no roots in $\mathbb{R}$, yet is not irreducible.

Lemma 1.16. *For a field $\mathbb{K}$, a polynomial $p(x) \in \mathbb{K}[x]$ with $2 \leq \deg p \leq 3$ is irreducible if and only if it has no roots in $\mathbb{K}$.*

Lemma 1.17 (Gauss). *Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ is primitive, that is $\gcd(a_0, a_1, \dots, a_n) = 1$. Then $p(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Lemma 1.18 (Eisenstein's Criterion). *Let $q(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ and suppose there exists a prime $p \in \mathbb{P}$ such that:*

1. $p \nmid a_n$
2. $p \mid a_0, \dots, a_{n-1}$
3. $p^2 \nmid a_0$

Then $q(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 1.19.

1. The polynomial

$$q(x) = 2x^3 + 6x^2 - 9x + 15$$

satisfies Eisenstein's criterion with $p = 3$.

2. The polynomial

$$q(x) = x^2 + x + 1$$

does not satisfy Eisenstein's criterion. Nevertheless, one can consider the transformation

$$r(x) = q(x + 1) = (x + 1)^2 + (x + 1) + 1 = x^2 + 3x + 3,$$

which does satisfy the criterion. Hence $q(x)$ is also irreducible.

More generally, using the same method as above, one can show that for any prime $p \in \mathbb{P}$, the polynomial

$$q(x) = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible.

Example 1.20. We recall the following basic homomorphisms:

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1. Suppose R is a ring. Then there exists a unique ring homomorphism

$$\varepsilon : \mathbb{Z} \rightarrow R, \quad n \mapsto n \cdot 1_R.$$

2. Suppose $\mathbb{F}$ is a field. Let $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ be the ring homomorphism from the above, and let $I = \ker \varepsilon$. Then ε induces a monomorphism $\mathbb{Z}/I \rightarrow \mathbb{F}$. In particular, $\mathbb{Z}/I$ is an integral domain, therefore $I \subseteq \mathbb{Z}$ is a prime ideal. Thus $I = \{0\}$ or $I = (p)$, for some $p \in \mathbb{P}$. This “formalizes” our previous definition of $\text{char } \mathbb{F}$.

- i. If $\text{char } \mathbb{F} = 0$, then $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ is a monomorphism. In particular, ε extends to a monomorphism $\mathbb{Q} \rightarrow \mathbb{F}$.

- ii. If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then $\text{ran } \varepsilon \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.

3. Suppose we have a fixed homomorphism of rings $\iota : R \rightarrow S$. Then for every choice of $\theta \in S$ there exists a unique ring homomorphism

$$\text{ev}_\theta : R[x] \rightarrow S$$

such that $\text{ev}_\theta(r) = \iota(r)$, for all $r \in R$, and $\text{ev}_\theta(x) = \theta$, namely the evaluation map $p(x) \mapsto p(\theta)$.

Definition 1.21. Let $\mathbb{F}$ be a field. A *(field) extension* of $\mathbb{F}$ is a field $\mathbb{K}$ containing^[2] $\mathbb{F}$ as a subfield. We will also write $\mathbb{F} \hookrightarrow \mathbb{K}$ to denote this relationship.

Example 1.22.

1. $\mathbb{Q} \hookrightarrow \mathbb{R}$
2. $\mathbb{R} \hookrightarrow \mathbb{C}$

If $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension, then $\mathbb{K}$ can be viewed as a vector space over $\mathbb{F}$.

Definition 1.23. The *degree* of a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is defined as the dimension of $\mathbb{K}$ as a vector space over $\mathbb{F}$. We also denote this as $[\mathbb{K} : \mathbb{F}]$ or $\dim_{\mathbb{F}} \mathbb{K}$.

Example 1.24.

^[2]Again, up to a monomorphism.

1. $[\mathbb{R} : \mathbb{Q}] = \infty$
2. $[\mathbb{C} : \mathbb{R}] = 2$

Lemma 1.25. Suppose we have field extensions $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$, then

$$[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{F}].$$

Definition 1.26. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $\alpha \in \mathbb{K}$. We say that α is *algebraic* over $\mathbb{F}$ if there exists a non-zero polynomial $p(x) \in \mathbb{F}[x]$ such that $p(\alpha) = 0$. Otherwise α is called *transcendental* over $\mathbb{F}$.

Example 1.27.

1. $\sqrt{2}, \sqrt{3}$ are algebraic over $\mathbb{Q}$.
2. π, e are transcendental over $\mathbb{Q}$.

Definition 1.28. We say that an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is *algebraic*, if every element $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Otherwise, we say that the extension is *transcendental*.

1.2. Minimal polynomials

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha \in \mathbb{K}$ is an algebraic element over $\mathbb{F}$. Then

$$I = \{p(x) \in \mathbb{F}[x] : p(\alpha) = 0\}$$

is a nonzero ideal in $\mathbb{F}[x]$. Since $\mathbb{F}[x]$ is a principle ideal domain, there exists a unique monic polynomial $m_{\alpha, \mathbb{F}}(x) \in \mathbb{F}[x]$, such that $I = (m_{\alpha, \mathbb{F}}(x))$.

Definition 1.29. The polynomial $m_{\alpha, \mathbb{F}}(x)$ in the above is called the *minimal polynomial* of α over $\mathbb{F}$. If $\mathbb{F}$ is clear from the context, we also abbreviate this as $m_{\alpha}(x)$.

Lemma 1.30. A monic polynomial $p(x) \in \mathbb{F}[x]$ is a minimal polynomial of $\alpha \in \mathbb{K}$ if and only if

1. $p(\alpha) = 0$, and
2. $p(x)$ is irreducible over $\mathbb{F}$.

Proof.

“ $\implies$ ”: Suppose $p(x) = m_{\alpha}(x)$. By definition, $p(\alpha) = 0$ and $p(x)$ generates the ideal of polynomials vanishing at α . Suppose $p(x) = q_1(x) \cdot q_2(x)$. Then $p(\alpha) = q_1(\alpha) \cdot q_2(\alpha)$.

Without loss of generality, assume that $q_1(\alpha) = 0$, that is $q_1(x) \in (p(x))$, therefore

$p(x) \mid q_1(x)$. But also $q_1(x) \mid p(x)$ by our assumption. Therefore $p(x)$ and $q_1(x)$ are associate in $\mathbb{F}[x]$, hence $q_2(x)$ is a unit.

“ $\Leftarrow$ ”: Similarly to the above. ■

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Since $m_{\alpha, \mathbb{F}}(x)$ is irreducible, the ideal $(m_{\alpha, \mathbb{F}}(x))$ is maximal. Hence $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x))$ is a field.

Proposition 1.31. *In the context above, $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) \cong \mathbb{F}(\alpha)$.*

Proof. Consider the evaluation homomorphism

$$\text{ev}_\alpha : \mathbb{F}[x] \rightarrow \mathbb{K}.$$

The image of ev_α is $\mathbb{F}[\alpha]$, the subring of $\mathbb{K}$ generated by $\mathbb{F}$ and α . But the image is also isomorphic to $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) = \mathbb{F}[x]/\ker \text{ev}_\alpha$, which is a field. ■

Corollary 1.32. *In the context above, $[\mathbb{F}(\alpha) : \mathbb{F}] = \deg m_{\alpha, \mathbb{F}}$.*

Proof. This follows from the general fact that, for a monic polynomial $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$, it holds that

$$\mathbb{F}[x]/(f(x)) \cong \{b_0 + b_1x + \dots + b_{n-1}x^{n-1} : b_0, \dots, b_{n-1} \in \mathbb{F}\} \cong \mathbb{F}^n. \quad \blacksquare$$

Note that conversely, if $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$ is an irreducible polynomial, then $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x))$ is a field extension of $\mathbb{F}$ of degree n . Furthermore, the element $x + (p(x)) \in \mathbb{F}[x]/(p(x))$ is a zero of $p(x)$, and $p(x)$ is the minimal polynomial of the element $x + (p(x))$.

Lemma 1.33. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension, then $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if $[\mathbb{F}(\alpha) : \mathbb{F}] < \infty$.*

Proof.

“ $\Rightarrow$ ”: Follows immediately from the previous corollary.

“ $\Leftarrow$ ”: The elements $1, \alpha, \alpha^2, \dots, \alpha^n$ are linearly dependant over $\mathbb{F}$. Therefore there exist $s_0, s_1, \dots, s_n \in \mathbb{F}$, not all zero, such that $s_0 + s_1\alpha + \dots + s_n\alpha^n = 0$, therefore α is algebraic over $\mathbb{F}$. ■

Corollary 1.34. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha, \beta \in \mathbb{K}$ are algebraic over $\mathbb{F}$. Then $\alpha \pm \beta, \alpha\beta, \frac{1}{\alpha}$ (provided that $\alpha \neq 0$) are algebraic over $\mathbb{F}$.*

Proof. By the previous lemma, it suffices to show that the respective generated extensions are of finite degree. The fields $\mathbb{F}(\alpha \pm \beta), \mathbb{F}(\alpha\beta)$ are subfields of $\mathbb{F}(\alpha, \beta)$. But the latter extension has finite degree, since

$$[\mathbb{F}(\alpha, \beta) : \mathbb{F}] = [\mathbb{F}(\alpha, \beta) : \mathbb{F}(\alpha)] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] \leq [\mathbb{F}(\beta) : \mathbb{F}] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] < \infty.$$

The final claim follows from the fact that $\mathbb{F}(\alpha) = \mathbb{F}(1/a)$. ■

Corollary 1.35. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$. Then the set of all algebraic elements of $\mathbb{K}$ is a sub-extension of $\mathbb{K}$, in other words it is a sub-field of $\mathbb{K}$ that is also an extension $\mathbb{F}$. This is called the algebraic closure of $\mathbb{F}$ in $\mathbb{K}$.*

Example 1.36. We want to determine the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$. We consider that we ultimately want something like $x = \sqrt{2} + \sqrt{3}$. Squaring yields $x^2 + 5 + 2\sqrt{6}$, or equivalently $x^2 - 5 = 2\sqrt{6}$. Squaring again yields $x^4 - 10x^2 + 1$. It remains to show that this polynomial, say $p(x)$, is minimal, which will be the case if we can show that it is irreducible over $\mathbb{Q}$.

Since $p(x)$ is monic with integer coefficients, by Gauss' lemma it suffices to rule out factorizations over $\mathbb{Z}$. Note that, by the rational root theorem, the only root candidates are ± 1 , and neither turn out to actually be roots. Hence there can be no linear factors.

Suppose that

$$\begin{aligned} p(x) &= x^4 - 10x^2 + 1 \\ &\stackrel{?}{=} (x^2 + ax + b)(x^2 + cx + d) \\ &= x^4 + ax^3 + bx^2 + cx^3 + acx^2 + bcx + dx^2 + adx + bd \\ &= x^4 + (a + c)x^3 + (b + ac + d)x^2 + (ad + bc)x + bd \end{aligned}$$

with $a, b, c, d \in \mathbb{Z}$. By comparing coefficients, we get

- (i) $a + c = 0$,
- (ii) $b + ac + d = -10$,
- (iii) $ad + bc = 0$,
- (iv) $bd = 1$.

From (i) we get $a = -c$, from (iii) we get $a(d - b) = 0$.

- If $a = 0$, then $c = 0$ and from (ii) we get $d = -10 - b$. From (iv) we get

$$1 \stackrel{?}{=} b(-10 - b) = -10b - b^2 \iff b^2 + 10b - 1 = 0,$$

which yields $b = -5 \pm \sqrt{96} = -5 \pm 2\sqrt{6} \notin \mathbb{Q}$.

- If $d = b$, then from (iv) we get $1 = b^2$, thus $b = \pm 1$, and from (ii) we get $a^2 - 2b = 10$. The case $b = 1$ yields $a^2 = 12$, while $b = -1$ yields $a^2 = 8$, both of which are not perfect squares.

Therefore we have shown that $p(x)$ is irreducible over $\mathbb{Q}$, and hence the minimal polynomial of $\sqrt{2} + \sqrt{3}$.